

Module 36 - Destructuring & Nested Destructuring

1. What is Destructuring?

Destructuring is a JavaScript feature that allows us to extract values from an array or properties from an object and store them in variables easily.

2. What is Nested Destructuring?

Nested Destructuring is a JavaScript feature that allows us to extract values from nested arrays or nested objects and store them in variables easily.

Example Code:

I. Array Destructuring

```
let arr = [10, 20, 30, 40];  
  
// Destructuring: Stores the 1st value in 'a' and the 4th value in 'd'.  
  
// Empty commas (,) skip the unwanted values.  
  
let [a, , , d] = arr;  
  
console.log(a, d);  
  
let arr1 = [10, 20, 30, 40, 5, 4, 5, 6, 7, 8, 9, 10];  
  
// '...a4' (Rest Operator) stores all remaining values in an array.  
  
let [a1, a2, a3, ...a4] = arr1;  
  
console.log(a1, a2, a3, a4);
```

II. Array Nested Destructuring

```
let nestArr = [1, 2, 3, 4, [10, 20, [30, 40]]];  
  
// Extracts values from the nested arrays directly into variables.  
  
let [A, B, C, D, [A1, A2, [B1, B2]]] = nestArr;  
  
console.log(A, B, C, D, A1, A2, B1, B2);
```

III. Object Destructuring

```
let student = {  
  name: "Rahul",  
  age: 22,  
  course: "MCA",  
  city: "Puducherry"  
};  
  
// Extracts only the 'name' and 'city' properties from the object.  
  
let { name, city } = student;  
console.log(name, city);  
  
let employee = {  
  empId: 101,  
  empName: "John",  
  department: "IT",  
  salary: 50000,  
  location: "Chennai"  
};  
  
// '...empDetails' stores the remaining properties in a new object.  
  
let { empId, empName, ...empDetails } = employee;  
console.log(empId, empName, empDetails);
```

IV. Object Nested Destructuring

```
let person = {  
  id: 1,  
  name1: "Devesh",  
  address: {  
    city1: "Puducherry",  
    state: "Tamil Nadu",  
    pin: {  
      code: 605110,  
      area: "Villianur"  
    }  
  }  
};  
  
// Extracts values directly from nested objects into variables.  
  
let {  
  id,  
  name1,  
  address: {  
    city1,  
    state,  
    pin: { code, area }  
  }  
} = person;  
  
console.log(id, name1, city1, state, code, area);
```